MIL-C-13678B(WC)
29 May 1973
SUPERSEDING
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MILITARY SPECIFICATION

CANNON, 37MM GUN, SUBCALIBER: M13

1. SCOPE

1.1 This specification covers one type of a subcaliber cannon to be inserted in the bore of a 105MM Howitzer Cannon M2A1 or M2A2 to enable the use of 37MM ammunition for training purposes.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-13931 - Cannons, Mortars, Recoilless Rifles and Recoilless Guns; General Specification for

MIL-P-14232 - Parts, Equipment and Tools for Army Materiel, Packaging and Packing of

MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-109 - Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army Weapons Command

B8769069 - Magnetic Particle Inspection of Cannon & Mount Components

PUBLICATIONS

U.S. Army Weapons Command

EPL 7238508 - Engineering Parts List for Cannon, 37MM Gun, Subcaliber: M13

FSC-1010

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U.S. Army Weapons Command cont'd.

IEL 7238508 - List of Inspection Equipment for Cannon, 37MM Gun, Subcaliber: M13

LSQAP 7238508 - List of Supplementary Quality Assurance Provisions for Cannon, 37MM Gun, Subcaliber: M13

PDS P7238508 - Packaging Data Sheet for Cannon, 37MM Gun Subcaliber: M13

(Copies of specifications, standards, drawings and publications, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials
ASTM-E138 - Wet Magnetic Particle Inspection

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 First article. When specified in the contract, the contractor shall submit a first article. Unless otherwise specified (see 6.1.1), the first article shall be included in the pilot pack (see 5.1).

3.2 Materials, design and construction. Subcaliber cannons and parts thereof shall conform to this specification, the drawings and specifications listed in Engineering Parts List EPL 7238508, and MIL-C-13931.

3.3 Breech mechanism. The breech mechanism of the howitzer shall perform the breechblock function for the subcaliber cannon when the cannon is inserted in the howitzer tube.

3.3.1 Firing mechanism operation. The firing mechanism of the howitzer shall fire the 37MM cartridge seated in the chamber of the subcaliber cannon.

3.3.2 Extractor operation. After firing, opening of the breech of the howitzer shall cause the extractor of the howitzer to engage the extractor of the subcaliber cannon to move the fired cartridge case rearward, permitting access for removal by hand.

3.4 Mechanical requirements.

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3.4.1 Deflector and tube assembly. The rear face of the deflector shall bear against the external shoulder on the muzzle end of the subcaliber cannon tube, when these parts are screwed together.

3.4.2 Extractor. The bore of the extractor of the subcaliber cannon shall be free from roughness or sharp edges. The extractor shall slide smoothly on the tube through its entire range of movement.

3.5 Performance.

3.5.1 High pressure resistance. The cannon shall be capable of withstanding 113 + 4 percent of 18,500 pounds per square inch upper pressure limit at 70°F when tested as specified in 4.5.2.4.4.

3.5.2 Material soundness. After testing, all parts shall be free of cracks and fractures and shall comply with any soundness requirements specified on the applicable drawings.

3.5.3 Bore enlargement. If proof firing is used to evaluate high pressure resistance, bore diameters, measured across the rifled lands after proof firing no more than 10 rounds, shall not exceed the basic (minimum) diameter specified on the applicable drawings by more than the amounts shown in Table I.

Table I Bore enlargement

Distance forward of Commencement of rifling	1/8 inch	1 1/2 inches	3 inches to muzzle
Amount (in inches)	0.005	0.005	0.004

NOTE: Interpolate to determine the bore enlargement between the distances given.

3.6 Workmanship. Workmanship requirements shall be in accordance with MIL-C-13931.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Quality assurance terms and definitions. Quality assurance terms and definitions shall be as specified in MIL-STD-109.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

4.4 First article inspection. A first article shall be submitted for inspection in accordance with the contract requirements. The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to quality conformance inspections specified herein and such other inspection as is necessary to determine compliance with the requirements of the contract.

4.5 Quality conformance inspection.

4.5.1 Examinations.

4.5.1.1 Inspection provisions. Cannons shall be inspected for conformance to requirements of the provisions of this specification, MIL-C-13931, SQAPs listed in LSQAP 7238508 and SQAP-APPENDIX-WVT.

4.5.1.2 Protective coatings. Examinations and tests of coatings shall be in conformance with the specifications referenced on the applicable drawings.

4.5.1.3 Workmanship. Parts shall be examined visually to determine compliance with the workmanship requirements of MIL-C-13931.

4.5.1.4 Material soundness. All parts shall be inspected visually prior to proof firing to determine freedom from cracks and fractures. In addition, each barrel and deflector shall be magnetic particle inspected in accordance with ASTM-E138. Any evidence of cracks or fractures shall be cause for rejection of the defective parts. Barrels which have inclusions exceeding Std IV B of B8769067 shall also be rejected.

4.5.1.5 Final examination. Each assembled subcaliber cannon shall be visually and manually examined for completeness and security of parts. Final examination of each subcaliber cannon shall be completed after testing and prior to preservation and packaging to determine compliance with this specification, MIL-C-13931 and as specified in the contract.

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4.5.1.6 Inspection of preparation for delivery. Unless otherwise specified, inspection to determine compliance with cleaning, preservation, packaging, packing and marking requirements of the applicable packaging data sheets for the level designated in the contract shall be as specified in MIL-P-14232.

4.5.2 Tests.

4.5.2.1 Materials. Materials used in the manufacture of the subcaliber cannons shall be tested in accordance with the material specifications referenced on the part drawings.

4.5.2.2 Magnetic particle inspection. Magnetic particle inspection, when specified on a drawing, shall be conducted in accordance with ASTM-E138.

4.5.2.3 Bearing surfaces. Surfaces of the deflector and tube assembly for which bearing is specified shall be visually examined to assure that the deflector bears against the external shoulder on the muzzle end of the subcaliber tube. (See 3.4.1).

4.5.2.4 Functioning tests.

4.5.2.4.1 Extractor. The extractor shall be manually moved through its entire range of movement to determine compliance with 3.4.2.

4.5.2.4.2 Extraction operation. Extraction shall be tested, with the subcaliber cannon inserted in the howitzer, using an empty service cartridge case to determine compliance with the requirements of 3.3.2.

4.5.2.4.3 Inspection at the proving ground.

4.5.2.4.3.1 Inspection procedures. Unless otherwise specified, the following inspections (examinations and tests) shall conform to the following applicable paragraphs in MIL-C-13931:

(a) Inspection before proof firing.

(b) Proof firing.

(c) Inspection after proof firing.

(d) Proof marking.

4.5.2.4.4 Proof firing. Each subcaliber cannon shall be fired in accordance with Table II. The standard weight projectile weighs 1.21 pounds. The upper pressure limit at 70°F is 18,500 pounds per square inch.

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Proof firing cont'd.

TABLE II FIRING SCHEDULE

No. of Rounds	Upper pressure Limit, %
1	113 ± 4

4.5.2.4.5 Proof firing results. If in the judgment of the Proof Firing Director, the results of the round specified in Table II are questionable or inconclusive, additional rounds having pressures not exceeding that in Table II may be fired.

4.5.2.5 Inspection after proof-firing. Unless otherwise specified, the inspection after proof-firing as specified in MIL-C-13931 shall apply.

4.5.2.5.1 Material soundness. After proof-firing, all parts shall be examined visually for cracks and fractures. Any evidence of cracks or fractures shall be cause for rejection. In addition, all tubes and deflectors shall be magnetic particle tested in accordance with ASTM-E138 for compliance with the soundness requirements specified on the applicable drawings.

4.5.2.5.2 Bore enlargement. After proof firing, no more than ten (10) rounds, the bore diameter across the lands shall be measured with a star gage, an air gage or other suitable instrument to determine compliance with 3.5.3 and Table I.

4.5.2.6 Records. Cannon and repair tubes that pass firing tests shall have applicable data entered on DA Form 2408-4, "Weapon Record Data" card by the proving ground. Cannon and repair tubes accepted on a proof sampling plan shall have applicable data recorded on DA Form 2408-4 by the contractor and approved by the Government representative.

4.5.3 Inspection equipment. Data lists containing equipment used in the inspection of parts, subassemblies, and assemblies are listed in IEL 7238508 List of Inspection Equipment Lists.

4.5.3.1 Acquisition, maintenance and disposition. Unless otherwise specified (see 6.1.1), responsibility for acquisition, maintenance and disposition of acceptance inspection and test equipment and for all other inspection equipment required to perform inspection prescribed by applicable specifications, shall be in accordance with MIL-I-45607.

4.5.3.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it shall be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.5.4 Inspection approval stamp. The application of the inspection approval stamp shall be as specified in MIL-C-13931.

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5. PREPARATION FOR DELIVERY

5.1 Pilot pack. On each contract, a pilot pack shall be forwarded in accordance with 3.1. Pilot packs for the cannon shall be packaged to the level of packaging specified in the contract and packed level C in accordance with the requirements of Packaging Data Sheet P7238508.

5.2 Preservation, packaging, and marking. The cannon shall be unit packaged, packed, and marked in accordance with the requirements of Packaging Data Sheet P7238508 for the level of protection specified in the contract (see 6.1).

5.3 Repair parts. Repair parts shall be unit packaged, packed, and marked in accordance with the requirements of the applicable packaging data sheet (see 6.1.1).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number and date of this specification

b. Requirements for submission of and shipping instructions for first article (see 3.1 and 4.4) and pilot pack. (see 3.1 and 5.1)

(1) The contract should designate the examinations and tests to be performed by the contractor and the examinations and tests to be performed by the Government.

(2) The contract should specify the sample size (i.e., the number of units) of the first article.

(3) The contract should designate the inspection and test data that is required to be furnished the Government when the contractor is required to perform either or both examinations and tests.

(4) The contract should designate at whose expense a retest may be performed.

(5) The contract should specify that the acceptable first article shall be delivered in accordance with the terms of the contract, or

(6) Any additional or extended examinations and tests beyond the scope of this specification should be specified and identified as a separate item of the contract.

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Procurement requirements cont'd.

c. The contract should specify under what conditions simulated firing may be performed, the test equipment available, and the procedures to be followed.

d. The levels of preservation, packaging, and packing required

e. The serial numbers for the cannons

f. Availability of inspection equipment from the Government

g. Extent of supplier's responsibility for Government-furnished and for supplier-required final inspection equipment.

h. List of inspection equipment, responsibility for acquisition, maintenance, and disposition thereof, if other than as specified (see 4.5.3.1)

6.1.2 Contract data requirements. Monthly reports of the results of final examination and performance testing shall be as specified for delivery on DD Form 1423 in the contract.

6.2 When warrented, the contract should specify the application of MIL-Q-9858 or MIL-I-45208, as appropriate, on the Management Control Systems Summary List DD Form 1660.

6.3 Unless otherwise specified, the contract should specify the application of MIL-I-45607 and MIL-C-45662 on the Management Control Systems Summary List, DD Form 1660.

6.4 Definition of pilot pack. A requirement in conjunction with a specification or contract for a first article in order to base approval of packages.

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